GeoLabel - Label Layout Specification

V1.00

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1. Revision

Version	Changes	Author
18.09.2020	Drafts of the Document	NEXTMA

2. Introduction

The document details the rules for the production of Labels and Barcodes used within the DPDgroup network including DPDgroup Business Units (BU) and partners.

The label template is common to all shipments.

This document forms part of the universal DPDgroup definition incorporating:

- Label Layout Specification
- GeoRouting Functional Specification

3. Parcel Labels for Shipments

Geolabel is the standard label for shipments within the DPDgroup network including DPDgroup Business Units (BU) and partners.

To create shipping label can use a lot of different ways.
For example:

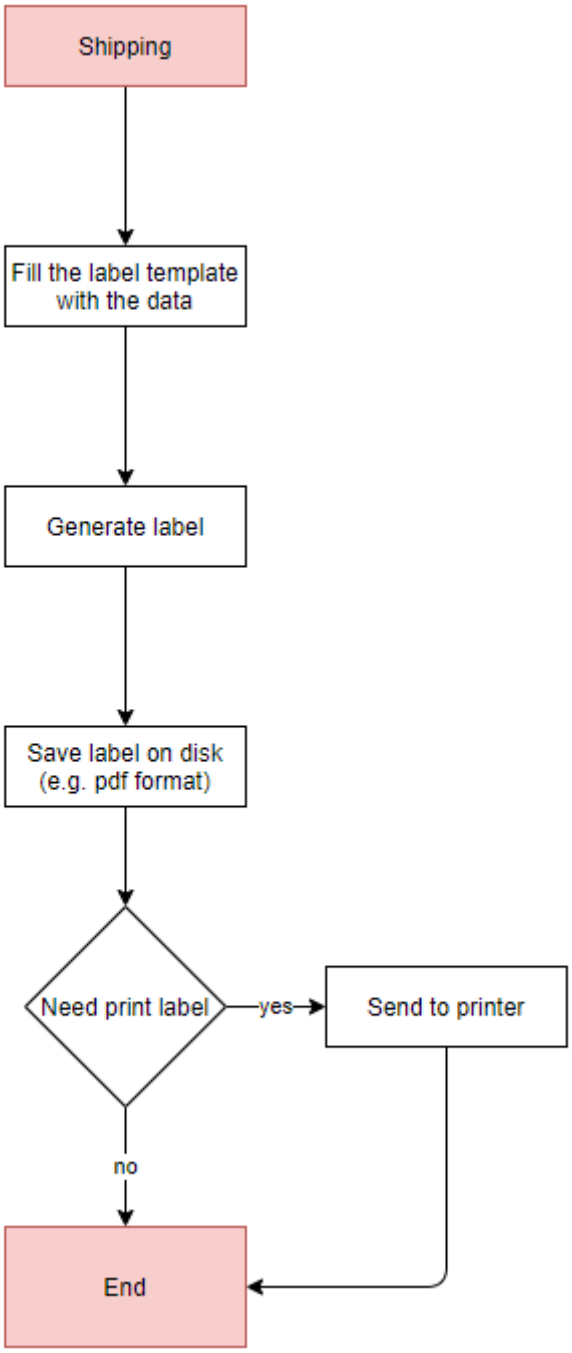
- html template
- library to create pdf
- manual drawing in programming language

The way of carrying out the task is free. It is necessary to provide the correct output file with dimensions and positions according to the label template.

The purpose of the primary bar code is to identify and route parcels within the DPDgroup network.

The purpose of secondary bar code is to provide additional parcel information to support a successful delivery and avoid errors in data entry. The secondary bar code provides a back-up solution in the event of no data at the point of scan such delivery options, network gateways and customs.
A second version of this barcode type (AZTEC) that include all parcel information.

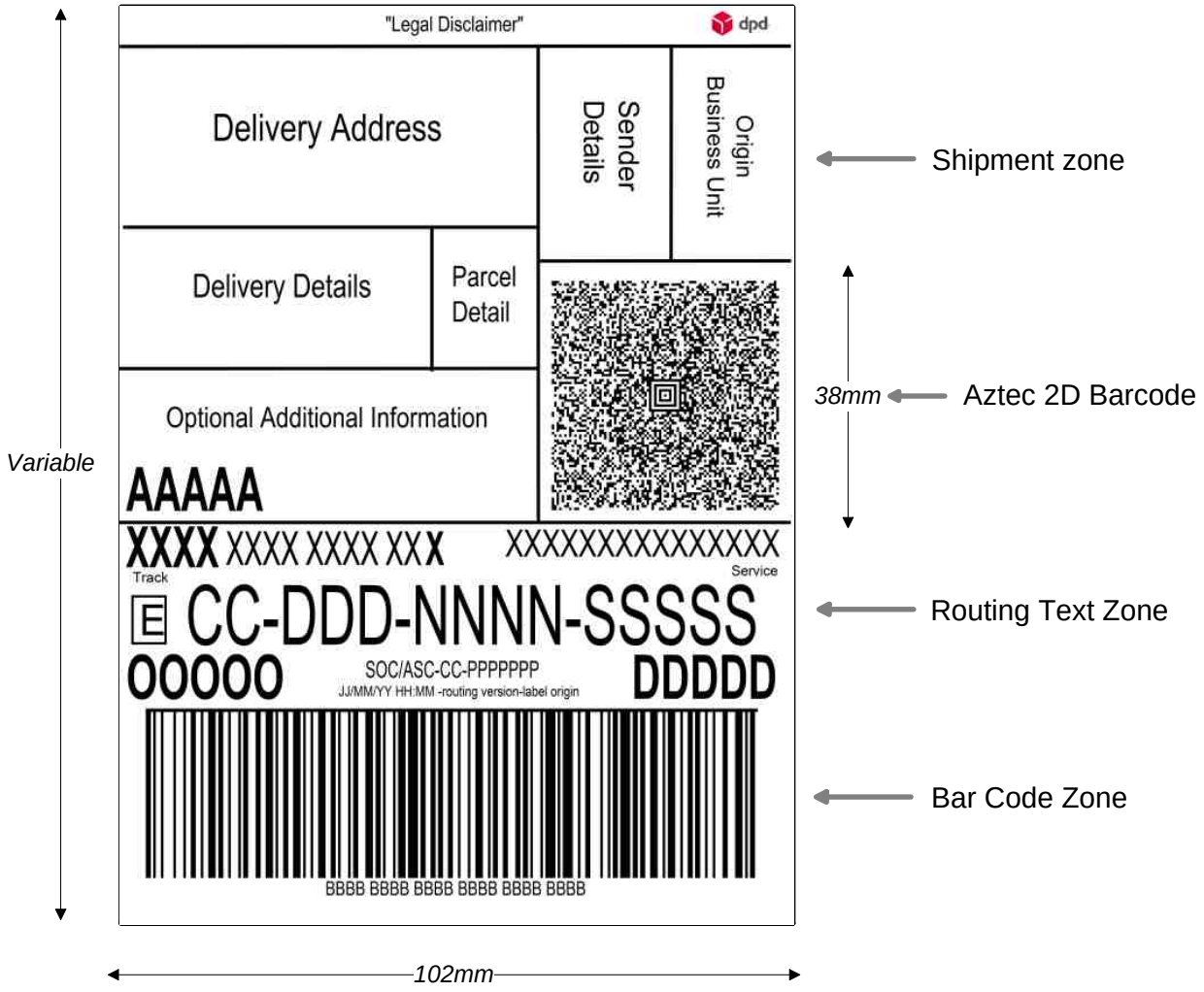
3.1. Create label example flow



3.2. Example

Example Home delivery Label

3.3. Label Zones



3.3.1. Label Zone Overview

Zone	Routing Label	Complete Label	Description
CODE 128 Bar Code	Mandatory	Mandatory	The Primary Bar Code to identify the package
Delivery Address	-	Mandatory	Package Receiver Address
Delivery Details	-	Mandatory	Package Receiver Details
Origin Business Unit	-	Mandatory	Origin Business Unit/Depot Details
Sender Details	-	Mandatory	Package Sender Details (or 'virtual' details, if private)
Routing Details	Mandatory	Mandatory	Text for parcel routing, the text will be retrieved from the DPDgroup Routing reference tables - determined by origin, destination, service, time of day.
Legal Disclaimer	-	Mandatory	Standard legal disclaimer text for intra DPDgroup shipments. Should be « Terms & Conditions of services on www.dpd.com ». This text can be doubled with a translation of it in the language of the origin business unit if there is enough space on the label. If not, only the English one should be printed on the label
Additional Info	-	Optional	
2D barcode		Conditional	It is mandatory to print the 2D barcode on the label to guaranty the good provision of the service payed by the customer. The lack of 2D barcode mean lack of backup solution of GeoData file

3.3.2. Routing Details

Routing text is printed on the label to enable human readable sorting; the following section illustrates the format and text with an example.

Note: Text sizes may vary dependent on printer and available fonts, the text proportions must be used to accommodate the mandatory text.

3.3.2.1. *Destination Text (D-Text)*

Information from DPDgroup reference files to enable manual sortation throughout delivery process.

Format: CC-DDD-NNNN-SSSSS
CC = Destination Country Code of delivery depot (ISO 2 digit alpha code)
DDD = Destination Business unit alpha code
NNNN = Destination Location in the destination Business unit (eg. Depot) – Numeric Representation

SSSSS = Destination business unit Sorting code

3.3.2.2. *Origin customer sorting Text (C-Sort)*

Information from DPDgroup reference files to enable manual sortation at customer location. To be filled according to origin Business unit's rules according to a specific agreement with customer.

Format: Up to 5 characters alphanumeric.
OOOOO = Origin Location sort identification

3.3.2.3. *Origin sorting Text (O-Sort)*

Information from DPDgroup reference files to enable manual sortation at origin Business Unit.

Format: Up to 5 characters alphanumeric.
OOOOO = Origin Location sort identification

3.3.2.4. *Destination location sorting Text (D-Sort)*

Information from DPDgroup reference files to enable manual sortation at destination location.

Format: Up to 5 characters alphanumeric.
DDDDD = Destination Location sort identification

3.3.2.5. *Destination location Micro-zoning sorting Text (M-Sort)*

Information from specific BU tools to enable manual sortation at destination location.
Could be in place of D-Sort on the sticker according to destination Business unit's rules.
This information will be printed by origin Business Unit only if there is a bi-lateral agreement with destination Business Unit.
This information is provided by specific Business unit's tools, provided by the destination Business unit.

Format: DDDDD
DDDDD = Destination Location micro-zoning sort identification

3.3.2.6. *Service Mark Identification*

Information to enable manual sortation of specific services. This is only required for certain services.

Format: E [Enclosed in a box]
E = Service Identification (X = Small Parcel, E = Express)

3.3.2.7. *Parcel Tracking No*

Format: **XXXX** XXXX XXXX XX**X** (15 Characters)
14 digit Parcel Identification number followed by a check digit.
Range Code (first 4 digits) must be highlighted in **bold**

3.3.2.8. *Service Text*

Information from DPDgroup reference files to enable visual identification of service

Format: XXXXXXXXXXXXXXXX (18 Characters)

3.3.2.9. *Service Code / Destination Postal Code*

Information to enable bar code reconstruction by manual re-input.

Format: SOC/ASC-CC-PPPPPPP
SOC = Sorting Code
ASC = Additional Service Code
CC = Country Code (ISO 2 digit numeric)
PPPPPPP= Postal Code

3.3.2.10. *Label Origin*

Format: 35 Characters

Information to identify the origin location of the label, the text must include the following details:-

- Date/Time the label was produced
- Routing Table version Number
- Origin Identifier (Customer/Depot/Workstation)
- Software used to produce the label

3.3.2.11. *Bar Code Text Representation*

Format: BBBB BBBB BBBB BBBB BBBB BBBB BBBB (28 Characters)

The bar code text printed under the bar code includes the last 27 digits of 28 digits printed in the bar code plus a check digit. (i.e. exclude the identification character “%”).

3.3.3. Label Field Sizes

Field Name	*Mandatory /Optional*	*Number of Characters*	*Text Height (20% Tolerance)*	*Comment*
ROUTING TEXT ZONE				
Destination Text (D-Text)	M	17	11mm **bold**	
Origin Routing Text (O-Sort)	M	5	7mm **bold**	In case of return thru PUDO, fix value “RET”
Destination Routing Text (D-Sort)	M	5	7mm **bold**	
Parcel Tracking No	M	15	6mm **bold** (4 digits) 4mm (10 digits) 4mm **bold** (1 digit)	
Service Mark Indicator	M	1	4mm	Character must appear in a box
Service Text	M	18	4mm	
Service Code / Country / Destination Postal Code	M	14	2mm	
Label Origin	M	35	1.5mm	
BAR CODE ZONE				
Bar Code Printer Line (Visual quality verification)	M	-	-	Appear above and for the length of the bar code
Code 128 Bar Code	M	-	25mm	
Bar Code Text	M	28	2mm	

The text size is dependent on the font capabilities of the label printer so a 20% tolerance is allowed. The label text size proportions must be maintained to ensure visual compatibility. The specification label examples use “Arial Narrow” font.

To distinguish between the letter “O” and the number “0” , the number should be printed with a diagonal line, if possible – i.e. “Ø”

3.3.4. CODE 128 Bar Code

The primary bar code used as part of the DPDgroup bar code strategy is a Code 128 bar code.

Code 128 is a variable length, high-density alphanumeric symbology. Three different subsets tell the bar code reader which character set to use initially. Three shift codes permit changing character sets inside a symbol. DPDgroup requires subset C for the numeric portion of the bar code. The alphanumeric portion of the bar code uses subset B.

Subset	***Description***
Code 128 A	The first sub-set encodes all uppercase and ASCII control characters
Code 128 B	The second sub-set encodes all upper case characters.
Code 128 C	The third sub-set encodes numeric digit pairs 00 through 99.

Each subset has its own advantages. For example if no alpha data is to be encoded the data density within the code can be effectively doubled by using subset C. Code 128 has 106 different bar and space patterns. Each pattern can have different meanings depending upon which subset is used.

3.3.4.1. *Bar-Code Data Content*

The data content of the DPDgroup Code 128 bar code is 28 characters as specified below:

Character		Subset	Field	Description
1	1	B	Identification Tag = "%"	DPDgroup barcode identification
2-8	7	B - Alpha C - Numeric	Destination Postal Code	Postal Code for the destination country
9-12	4	B – Alpha * C - Numeric	Origin Identification	Fixed Code to identify the DPDgroup Origin. The code will identify the origin location (eg. depot number).
13-22	10	C	Origin Parcel Number	Parcel Number linked to DPDgroup Origin
23-25	3	C	Service Code or Sorting Code	Intra-DPDgroup Service Code or Sorting Code
26-28	3	C	Destination Country Code	Extended ISO 3 digit Numeric Unicode of destination country or destination network (routing output)

NOTE: The shaded area indicates the unique DPDgroup 14 digit Tracking number.

* The numeric Tracking number has an exception for intra-France shipments (the UPU number) with the prefix containing an alphanumeric code.

The DPDgroup group tracking number has 14 digits; this number is used within the barcode and data interfaces across the group. For presentation on documents and reports the tracking number must be shown with 15 digits, the extra digit is the check digit to provide number integrity on entry validation.

3.3.4.1.1. *28 digit bar code structure*

Digit	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*	*14*	*15*	*16*	*17*	*18*	*19*	*20*	*21*	*22*	*23*	*24*	*25*	*26*	*27*	*28*
A/N	A	A	A	A	A	A	A	A	N*	N*	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N
Field	%	P	P	P	P	P	P	P	T	T	T	T	T	T	T	T	T	T	T	T	T	T	S	S	S	C	C	C

- Character Type (A/N)

A = Alphanumeric
N = Numeric
* = An exception to standard rule within France only (see below)

- Character Contents (Field)

% = DPDgroup identification Tag

P = Destination Postal Code (7AN or 7N format)
If the Postal Code is shorter the 7 digits then the field is padded with “0” to the left.
Examples:
56349 = 0056349
B61AA = 00B61AA

T = Tracking Number (DPDgroup)
Origin Location or depot number (4N format)
Origin Parcel Number (10N format)

= Tracking Number (Intra-France UPU Number)
Prefix (2AN format)
Origin Parcel Number (9N format)
Suffix (3N format)

Note: A tracking number is unique for a period of 6 months before re-use.

S = Service Code / Sorting Code

C = Destination Country Code / Network code (ISO 3-digit Numeric)

3.3.4.1.2. *Check Digit Calculation Algorithms*

- Code 128 Bar Code

Check digits are included in the human readable interpretation of the DPDgroup code 128 bar code, this is designed to prevent errors for manual data entry. These check digits should not be confused with the internal code 128 symbology check digit. The symbology check digit is a required part of the code 128 symbol structure and is not counted as part of the user defined data encoded in the symbol. This check character is necessary for the bar code reader to ensure correct decode performance and is usually calculated by the bar code printing application.

Check digits are required for Human Representation of:

- 28 Digit bar Code
- 14 Digit Parcel Identification (Origin Location plus Origin Parcel Number)

- Check Digit Calculation Algorithm

Check Digits will be generated using ISO/IEC 7064 mod 37/36 standard, the following table indicates the value to be used within the algorithm for each alphanumeric character of the field to calculate the check digit.

- Character Table for ISO 7064 mod 37/36

Char	*Value*	*Char*	*Value*	*Char*	*Value*
0	0	A	10	N	23
1	1	B	11	O	24
2	2	C	12	P	25
3	3	D	13	Q	26
4	4	E	14	R	27
5	5	F	15	S	28
6	6	G	16	T	29
7	7	H	17	U	30
8	8	I	18	V	31
9	9	J	19	W	32
		K	20	X	33
		L	21	Y	34
		M	22	Z	35

3.3.4.2. *Physical Properties*

The specified physical properties of the DPDgroup bar code have been developed to ensure accurate readability with all bar code scanning systems in use and in the foreseeable future.

The individual components of the specification were determined as follows:

3.3.4.2.1. *Bar / Space Width*

Determined by the scanner type, depth of field and presentation speed.

3.3.4.2.2. *X-Dimension*

The width of the minimum narrow element (bar or space) is a dimensional parameter, which can be used to calculate the symbol's actual width

3.3.4.2.3. *Bar / Space Tolerance*

This is the amount the bars and/or spaces can vary from symbol to symbol without changing the overall print quality grade or significantly altering the size of the symbol. The tolerance is roughly equivalent to 20% of the X-dimension.

3.3.4.2.4. *Horizontal Label Registration Tolerance*

How much the different colors of ink used to print the bar code label can be out of alignment with one another. This is primarily related to multi-color offset printing.

3.3.4.2.5. *Width*

The overall width of the bar code is a result of the X-dimension and wide to narrow ratio. The width usually cannot be set independently.

3.3.4.2.6. *Height*

The overall height of the bar code was established to ensure that bar codes were not so small that they could pass between two scanners without being read. Maximum height is significant as increased height improves scan capability.

3.3.4.2.7. *Quiet Zone*

The specified Quiet Zone is a requirement taken from the Code 128 symbology specification and is equivalent to 10 times the maximum X-dimension. A 1mm quiet zone shall also be applied above and below the symbol.

3.3.4.2.8. *Human Readable Text*

The human readable text is required by DPDgroup and is used to enable an operator to manually enter the data encoded in the bar code in case of scanner failure or label damage. Extra human readable information is also included to enable the operator to determine the delivery service and origin postal code (required for returns).

3.3.4.2.9. *Bar Code Metrics*

The important physical properties evaluated on the DPDgroup Code 128 bar code are:

THERMAL PRINTED

Bar Size:
X-Dimension (Narrow Bar Width)

Minimum:	0.375mm
Maximum:	0.4mm

Overall Bar Code Size:
Dimensions of thermally printed bar codes

Min 80mm x 25mm height (Numeric Business Unit)
Max 91.5mm x 25mm height (Alpha Business Unit)

Label Stock:
Use only white label stock. (Direct thermal printing preferred)

LASER PRINTED

Bar Size:
X-Dimension (Narrow Bar Width)

Minimum:	0.375mm
Maximum:	0.5mm

Bar Code Size:
Dimensions of laser printed bar codes

Min 80mm x 25mm height (Numeric Business Unit)
Max 115mm x 25mm height (Alpha Business Unit)

3.3.4.2.10. *Bar-Code Quiet Zones*

10.0 times the X-Dimension on each side (5mm)
5mm on start and end of bar code extending at 45 degrees from the corners (see below)
1mm on top and bottom of bars

3.3.4.2.11. *Bar Code Size (Intra DPDgroup shipments)*

The following Table indicates the printed 28-digit Bar-code size between countries assuming the minimum narrow bar width (0.375mm): -

From\To	GB	FR	DE	Other
GB	91.5mm	80mm	80mm	80-91.5mm
FR	91.5mm	*93mm	80mm	80-91.5mm
DE	91.5mm	80mm	80mm	80-91.5mm
Other	91.5mm	80mm	80mm	80-91.5mm

The bar-code size is dependent on the Printers capability to print Code 128 with differing subsets B & C.

* Larger size for intra France shipments exceeds maximum size to accommodate an alphanumeric tracking number

3.3.4.3. *Print Quality*

The print quality specifications were developed to ensure all scanners could read the bar codes. The major print quality criteria are:

3.3.4.3.1. *Print Contrast Signal (PCS)*

The Print Contrast Signal values were established to ensure the bar codes would be readable at the conveyor speeds used on the DPDgroup automated package sorters. The scanners cannot consistently read bar codes with low PCS values on the sorters, although they may read well with a hand scanner.

High quality white label stock along with high quality print is required to achieve the specified PCS. (Thermal labels should be of a Near Infra-Red type to support scanners operating in the 630nm to 670nm range. Labels should also exhibit stable print characteristics through exposure to varying heat and temperature conditions.)

3.3.4.3.2. *Decode Margin*

Decode margin is a graded measure of de-codability; or how close a given scan comes to a reference decode failure. Lower margins indicate an increased susceptibility to decoding failure due to scanning errors.

3.3.4.3.3. *Average Bar Error*

The average bar growth in the symbol relative to its X-dimension.

3.3.4.3.4. *Reflectance*

Given in two parts, reflectance is a measure of the reflected laser light from the bars (dark) and spaces (light).

3.3.4.3.5. *Defects*

This is a graded measure of the maximum element reflectance non-uniformity due to any noise within the element due to voids, spots, or fuzzy edges.

3.3.4.3.6. *Overall Print Quality*

Print Contrast Signal:

Minimum Print Contrast:	90%
Desirable Contrast Range:	95 - 100%

Symbol Contrast (min):	65%
Decode Margin (min):	“A” grade
Average Bar Error (max):	±0.10X
Defects (max):	15%

Reflectance:

Light:	>=80%
Dark:	<=15%

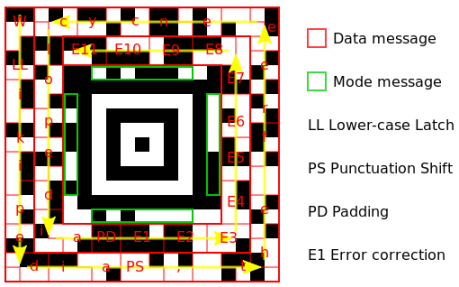
3.3.5. 2D-Barcode

Use of AZTEC Two Dimensional Symbol enables DPDgroup customers to encode large amounts of data into the symbol, which will travel with the package.

The symbology is capable of encoding the entire 255-character ASCII data set. The symbology can encode up to 3067 characters in a single symbol, however this is reduced by the error correction level and the application. DPDgroup has set a limit of 1000 characters. For a maximum square size of 3,4cm.

AZTEC Code has been invented by Andrew Longacre, Jr. and Robert Hussey in 1995. It has been released to the public domain.
This symbology use square modules with a unique finder pattern in the middle of the symbol.

The symbol is built on a square grid with a bulls-eye pattern at its center for locating the code. Data is encoded in concentric square rings around the bulls-eye pattern. Because of that, a quiet zone is not required.



Message data is placed in a spiral pattern around the core. The mode message begins "01011100", indicating 01_2+1 2 layers, and 011100_2+1 = 29 data codewords (of 6 bits each).

The selected encoding characters set is the ISO/IEC 8859-1 Latin alphabet No. 1

According to ISO 24778 (AZTEC bar code), the error correction is set to 23%.

3.3.5.1. *Data Content*

To insure compatibility with the market, and facilitate exchange with external providers, custom and security, the content of the barcode will respect the ISO 15434 norm.

This International Standard defines the manner in which data is transferred to high-capacity automatic data capture (ADC) media from a supplier's information system and the manner in which data is transferred to the recipient's information system. This International Standard does not define the internal data storage format for specific high-capacity ADC media. This International Standard does not specify the application of data structures provided by a specific data syntax format. The application of the data structure is specified by industry conventions.

Users of ADC technologies benefit by being able to receive data in a standard form and by being able to provide data in a standard form. Static ADC technologies such as bar code symbologies, magnetic stripe, optical character recognition, surface acoustical wave (SAW) and Weigand effect typically encode a single field of data. Most applications of these technologies involve the encoding of a single field of data by the supplier of the medium and the subsequent decoding of the data field by the recipient. Encoding single fields of data permits the supplier to perform the encodation from a single field within the supplier's information system. Decoding single fields of data permits the recipient to input this data into a single field in the recipient's information system, in lieu of key entry.

High-capacity ADC technologies, such as two-dimensional symbols, RFID transponders, contact memories and smart cards, encode multiple fields of data. These multiple fields are usually parsed by the recipient's information system and then mapped to specific fields of data in the recipient's information system. This International Standard defines the syntax for high-capacity ADC media, so as to enable ADC users to utilize a single mapping utility, regardless of which high-capacity ADC medium is employed.

3.3.5.2. *Special characters used by ISO 15434*

Where "$^{R}_{S}$", "$^{G}_{S}$", "$^{U}_{S}$", "$^{E}_{OT}$" are non-printable characters"

		ASCII Value	
Description	***Tag***	***decimal***	***Hexadecimal***
Format Envelope trailer	$^{R}_{S}$	30	0x1E
Field separator	$^{G}_{S}$	29	0x1D
Sub field separator	$^{U}_{S}$	31	0x1F
Message separator	$^{E}_{OT}$	04	0x04

3.3.5.3. *ISO message structure*

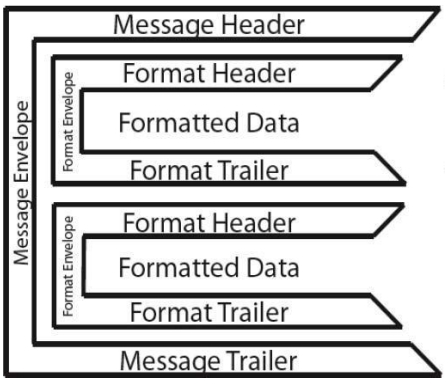
To allow multiple data Formats to be contained within a data stream, a two level structure of enveloping is employed. The outermost layer of the message is a Message Envelope that defines the beginning and end of the message. Within the Message Envelope are one or more Format Envelopes that contain the data (see Figure 1). Multiple formats in a single message should only be employed with bi-lateral agreements of the trading partners.

The Message Envelope shall consist of

- a Message Header,
- one or more Format Envelope(s), and
- a Message Trailer (when required).

Each Format Envelope within the Message Envelope shall consist of

- a Format Header,
- data, formatted according to the rules defined for that Format, and
- a Format Trailer (when required).



Description	Value	Comment
Message header	[)>$^{R}_{S}$	Beginning of the barcode
Format header	01 $^{G}_{S}$ 02	“01” for transportation and “02” the version
ISO Fields	Variable	Fields specified by the ISO norm
Format trailer	$^{R}_{S}$	
Format header	07	“07” for free definition of format (DPDgroup definition)
DPDgroup Header	G*NN*	To identify the type of DPDgroup format (from 02)
DPDgroup Fields	Variable	Fields depending on the DPDgroup type format
Format trailer	$^{R}_{S}$	
Format header	07	“07” for free definition of format (DPDgroup definition)
DPDgroup Header	Variable	To identify the type of DPDgroup format
DPDgroup Fields	Variable	Fields depending on the DPDgroup type format
Format trailer	$^{R}_{S}$	
….		
Message trailer	$^{E}_{OT}$	

3.3.5.4. *DPDgroup message structure*

Four class of DPDgroup format has been defined.
For each of those different classes, different format could be defined.

Description	Value	Comment
ISO format	$01^{G}{}_{s}02$	ISO 35434 fixe format. “01” for transportation block and “02” the version of this block.
DPDgroup basic format	$07^{G}{}_{s}$**G***NN*	Where “*NN*” is the version. Set to “03” in this version. It is a mandatory block that has to be just after the ISO format block.
Standard DPDgroup format	$07^{G}{}_{s}$**S***NNX*	Where: • “*NN*” defines the type of the format block. • “X” defines its versions. Those blocks are optional. Those blocks are defined by DPDgroup. Those blocks have to be placed behind the DPDgroup Basic format block. No order is defined between Standard DPDgroup blocks.
Intra-Business unit format	$07^{G}{}_{s}$**D***YYYNNX*	Where: • “*YYY*” defines the business unit (defined into BIC7 table). • “*NN*” defines the type of the format block for the BU. • “X” defines the version of this block. Those blocks are optional. Those blocks are defined per each business unit but this definition has to be given to DPDgroup for information. Those blocks have to be placed behind the Standard DPDgroup blocks if there is some. If not, those blocks have to be placed behind the DPDgroup Basic format block.
Customer format	$07^{G}{}_{s}$**C***A…A*	Where “A…A” has to be defined with DPDgroup. Those blocks are reserved for potential customer need.

3.3.5.4.1. *Standard DPDgroup format*

Description	Value	Comment
Sender format block	S01	To carry sender information (name, company, address, phone numbers…
COD format block	S02	To carry cash on delivery information (amount, currency, …)
Custom format block	S03	To carry information mandatory for custom process (parcel description, value, custom classification, VAT/EORI, commercial invoice information, …)
Return format block	S04	To carry information mandatory for the parcel return.
Digital Label block	S05	To carry information necessary for digital label. Specify as well that this 2D barcode is a digital one. Mustn't be inside a 2D barcode printed on a label.
Fresh/Food	S06	To carry information necessary for food and fresh
Fixed time delivery	S07	To carry information necessary for fixed time delivery service
GPS localization	S08	To carry consignee GPS localization

3.3.5.5. *Column description for format block description*

Column name	Description
Field No	Field or sub-field number. For field identification.
Description	Field description.
Max Size	Field maximum size. For sender, receiver and article information, this is the medium size of all its subfields.
Subfield Max Size	Subfield maximum size
Priority	Priority is between 1 and 10. Only fields of priority one are mandatory. This priority is also used to build the 2D barcode. If the total of the real size of all fields and subfields is more than 1000 characters. Then fields of priority 10 are suppressed (emptied). If the total of the real size of all remaining fields is still more than 1000 characters. Then fields of priority 9 are suppressed (emptied). ... When the total of the real size of all remaining fields is less than 1000 characters barcode is printed. Priority is set to "X" when the field priority is driven by a management rule.
Field / Sub Filed	"F" for field, "S" for subfield.
Mgt Rul	Number of management rule(s) that apply to this field (describe in "Mgt Rules" table)

3.3.5.6. *Management rules for the feeding of fields into format block description*

Mgt Rul	Description
2	The maximum size of the field is not equal to the sum of the maximum size of each sub-filed. It is a medium value of the total of each sub-field' medium size. If the maximum size of the field is reached when all subfield real size are added, we have to suppress subfield per order of priority or trunk subfields in the "best way". If there is some available space, the total size of the field can be up to the total of each sub-field' max size.
3	The sender block is mandatory into 2D barcode if the parcel is a non EU parcel.
4	At least one of the field "Receiver contact name" (2DSTD_RecContact) and "Receiver company name1" (2DISO_RecCompName1) is required (mandatory)
5	At least one of those fields is required (mandatory) for Predict. If one of those fields is set and if it is valid for this service code, the customer will have to receive a predict notification whatever is the value of "Notification type" (2DSTD_NotifType).
6	Most of the time, the field will be included into "Receiver street" (2DISO_RecStreet)
7	Mandatory for some countries (US/CA/ES...)
8	Mandatory if the parcel Is coming from EU to a non EU country, from a non EU country to EU country or non EU to non EU country. Empty for intra EU parcel.
9	Mandatory for limited quantity parcel, empty in the other case.
10	At least one of those two fields is mandatory
11	Mandatory if custom process
12	The type of the information will depend on the destination country
13	The type of the information will depend on the sender country
14	mandatory only if different to receivers detail in "DPDgroup basic format block"
15	mandatory only if different to "Destination Country Code" (2DISO_DestCountryCode)
16	If it is a MPS shipment, the custom block will be only set for the "Master parcel" ("2DISO_RangInNumber" = 001/NNN). Custom block is forbidden for the other piece of the MPS shipment ("2DISO_RangInNumber" <> 001/NNN).
17	Sender, receiver and shipment information (Shipment-ID, number of parcels) has to be already available in the 2D barcode. If it is a MPS shipment all piece must be shipped.
18	An article is described only if all field of the article description can be put into the barcode.
19	Total weight of consignment in case of MPS. The field is set only for the master piece of MPS. Empty in the other cases.
20	By default, fields are of variable length up to the maximum length defined. As well for numeric field than for alphabetic field, no padding has to be added to the data. There is some exception for which format is described into column "additional description".
21	If the field is of the type "F" (field) it has to be followed by the character ASCII 29 : <$^{G}_{S}$>
22	If the field is of the type "S" (subfield) it has to be followed by the character ASCII 31 : <$^{U}_{S}$>
23	The data in the barcode has to be closed by the character <$^{E}_{OT}$> ASCII 04

Mgt Rul	Description
24	Filed content has to be IOS 646 compliant
25	When possible, the phone number has to be set if the parcel is shipped to Ireland (or other country without zip code)
26	This field is mandatory if it's a predict parcel
27	Can be used to personalize the notification sent to the receiver
28	If the country has no zip code system, the zip code is set to one "0"
29	Mandatory if it is a "COD" parcel
30	If all subfields of a field are empty, subfield separators are not mandatory.
31	If a field or a subfield is empty, it mustn't be padded. The field or subfield separator has to be there and followed immediately by the previous one.
32	The field may contain Sorting code or Service code for compatibility reason. BIC7 Service code are in fact SOCode without ASCode. GeoRouting reference table will help in the identification of SOCode that have an associated ASCode and SOCode without ASCode (See GeoRouting specification)
33	A specific service element.will be set ASCode description in GeoRouting reference table when there will be an ASCode associated to the SOCode.
34	If the field "2DDLA_Paid" is set to "B" (to be bill). In this case, the filed "Customer Account Number" of the "ISO block" has to be set with an existing account number.
35	Only set if the collecting BU accept this billing method
36	Numeric field of variable length. Mustn't be padded
37	Date format DDMMYYYY DayMonthYear
38	Hour format HH:MM, Hour:Minute
39	Mandatory if Sending BU request to send notification when collecting parcel in PUDO Point

3.3.5.7. Block description

3.3.5.7.1. *ISO format block*

Field No	Name	Description	Max Size	Subfield Max Size	Type	Priority	Field / Sub Field	Mgt Rul	Additional Description
		Message Header	4		A	1			[)>RS
1	2DISO_Header	Format Envelope Header	2		N	1	F		"01" = Transportation
2	2DISO_Version	Format Version	2		N	1			"03" = Version
3	2DISO_DestZipCode	Destination Postal Code	9		AN	1	F	28	
4	2DISO_DestCountryCode	Destination Country / Network Code	3		N	1	F		ISO 3166 (eg. 826) or Network Code (900 range)
5	2DISO_SOCode	Service code / Sorting Code	3		AN	1	F	32	DPDgroup Service Code / DPDgroup Sorting Code
6	2DISO_ParcelNumber	Parcel Number	14		AN	1	F		Digit 1-4 = Business Unit Origin (or Depot) / Tracking number
7	2DISO_SCAC	Origin Carrier SCAC	4		AN	1	F		Standard Carrier Alpha Code ("GEOP")
8	2DISO_CustAccNumber	Customer Account Number	17		AN	10	F		Recognized Account number within the Origin Business Unit / Carrier assigned shipper ID
9	2DISO_JDPickup	Julian day of pickup	3		N	1	F		Numeric (padded on the left with zero). Julian day is presenting the interval of time in days and fractions of a day since January the 1st. Example: The julian day of 12.03.2012 is (31+29+12) = 72
10	2DISO_ConsCustRef1	Consignment customer reference number 1	35		AN	5	F		Consignment reference could be different from parcel reference / Shipment ID Number (assign by shipper)
11	2DISO_RangInNumber	Parcel X / in consignment Y (MPS)	7		AN	1	F		3N/3N (Padded on the left with zero) Default 001/001
12	2DISO_DeclaredWeight	Declared weight	7		AN	1	F		"nn.nnKG"
13	2DISO_CrossMatch	Cross match	1		A	1	F		"Y"/"N" Default "N"
14	2DISO_RecStreet	Receiver Street	35		AN	1	F		Could include the receiver Property Number. In this case Property number will be empty
15	2DISO_RecTown	Receiver Town/City	35		AN	1	F		
16	2DISO_RecState	Receiver State	2		AN	X	F	7	"CA" for California
17	2DISO_RecCompName1	Receiver Company Name1	35		AN	X	F	4	

Field No	Name	Description	Max Size	Subfield Max Size	Type	Priority	Field / Sub Field	Mgt Rul	Additional Description
		Format Envelope trailer	1		A	1			$^{R}_{S}$

3.3.5.7.2. *DPDgroup basic format block*

Field No	Name	Description	Max Size	Subfield Max Size	Type	Priority	Field / Sub Field	Mgt Rul	Additional Description
1	2DSTD_Header2	Format Envelope Header	2		N	1	F		"07"
2	2DSTD_FormatID	2D Format Identification	3		AN	1	F		Standard block Version "G03"
3	2DSTD_BarcodeOF	Barcode overflow	1		N	1	F		Flag (1 = yes / 0 = no) if limit of barcode size is exceeded
4	2DSTD_NoHandWoutData	No handling without data	1		N	2	F		Flag (1 = yes / 0 = no)
5	2DSTD_RoutingNec	Routing necessary	1		N	1	F		Flag (1 = yes / 0 = no)
		Receiver Information	150	343		1		2 30	Note: The receiver Postal Code/Country is contained within the primary bar code
6	2DSTD_RecComment	- *Receiver comment*		70	AN	5	S		Additional text to support delivery process. Any comment to help the delivery of the parcel. (Recipient comment)
7	2DSTD_RecCompName2	- *Receiver Company Name2*		35	AN	10	S		Could be the company department (Communication department, finance...)
8	2DSTD_RecContact	- *Receiver contact name*		35	AN	X	S	4	(receiver/consignee)
9	2DSTD_RecContactPho1	- *Receiver Contact Phone Number1*		25	AN	X	S	25	The only acceptable alphabetic character Is "+" before country code. Country code can be preceded of "00" or "+"
10	2DSTD_RecContactPho2	- *Receiver Contact Phone Number2*		25	AN	10	S		The only acceptable alphabetic character Is "+" before country code. Country code can be preceded of "00" or "+"
11	2DSTD_RecNotifMob	- *Receiver Notification Mobile Phone number*		25	AN	X	S	5	The only acceptable alphabetic character Is "+" before country code. Country code can be preceded of "00" or "+"
12	2DSTD_RecNotifEMail	- *Receiver Notification email*		50	AN	X	S	5	

Field No	Name	Description	Max Size	Subfield Max Size	Type	Priority	Field / Sub Field	Mgt Rul	Additional Description
13	2DSTD_RecPropNum	- Receiver Property Number / House No		8	AN	2	S	6	
14	2DSTD_RecAdd2	- *Receiver Address Line 2*		35	AN	2	S		
15	2DSTD_RecAdd3	- *Receiver Address Line 3*		35	AN	2	F		
16	2DSTD_NotifType	Notification type	1		A	X	F	26	Notification could be driven by this field or by the presence of the field notification phone/email (see rule number 5). "B" : Email+SMS, "E" : Email, "S" : SMS, "C" : Call, Empty : No notification.
17	2DSTD_TotalWeight	Total weight of shipment (MPS)	8		AN	1	F	19	"nnn.nnKG","nnnn.nKG","nnnnnnKG"
18	2DSTD_NotifSenderComp	Notification Sender Company Name1	35		AN	X	F	27	Refer to the friendly name of the company
19	2DSTD_NotifSenderContact	Notification Sender contact person	35		AN	X	F	27	
20	2DSTD_SendParcelRef	Sender parcel reference	35		AN	5	F		
21	2DSTD_RecParcelRef	Receiver parcel reference	35		AN	5	F		
22	2DSTD_ConsType	Consignment Type	1		AN	X	F	8	("D"/"N")Documents (no value) or Non-Documents (with value)
23	2DSTD_ContDescr	Contents Description	50		AN	X	F	8	
24	2DSTD_ConsCustRef2	Consignment customer reference number 2	35		AN	5	F		
25	2DSTD_LimitedQtyHaz	limited quantities hazardous	1		N	X	F	9	limited quantities hazardous goods parcel value (1 = yes / 0 = no), default: 0
26	2DSTD_ASCode	Additional service code	6		AN	1	F	33	Three characters to start then up to 6 characters (RFU)
		Format Envelope trailer	1		A	1			$^{R}{}_{S}$

3.3.5.7.2.1. Sender format block

Field No	Name	Description	Max Size	Subfield Max Size	Type	Priority	Field / Sub Field	Mgt Rul	Additional Description
1	2DS01_Header	Format Envelope Header	2		N	1	F		"07"
2	2DS01_FormatID	BLOCL-ID = "S01X"	4		AN	X	F	3	X (AN) is the version of the record: "1".
		Sender Information	140	270		1		2 30	
3	2DS01_SendCompName	- *Sender Company Name*		35	AN	X	S	10	
4	2DS01_SendPhone	- *Sender Phone Number*		25	AN	2	S		The only acceptable alphabetic character Is "+" before country code. Country code can be preceded of "00" or "+"
5	2DS01_SendNotifEMail	- *Sender Notification email*		50	AN	X	S	39	
6	2DS01_Contact	- *Sender Contact Name*		35	AN	X	S	10	
7	2DS01_SendPropNum	- *Sender Property Number / House No*		8	AN	2	S	6 11	
8	2DS01_SendStreet	- *Sender Street*		35	AN	1	S	11	
9	2DS01_SendAddr2	- *Sender Address Line 2*		35	AN	2	S	11	
10	2DS01_SendTown	- *Sender Town/City*		35	AN	1	S	11	
11	2DS01_SendZipCode	- *Sender Zip code*		9	AN	2	S	28 11	
12	2DS01_SendCountryCode	- *Sender Country code*		3	N	1	F	11	ISO 3166 (eg. 826 for GB)
		Format Envelope trailer	1		A	1			$^{R}{}_{S}$

3.3.5.7.3. *COD format block*

Field No	Name	Description	Max Size	Subfield Max Size	Type	Priority	Field / Sub Field	Mgt Rul	Additional Description
1	2DS02_Header	Format Envelope Header	2		N	1	F		"07"
2	2DS02_FormatID	BLOCL-ID = "S02X"	4		AN	X	F	29	X (AN) is the version of the record from "0".
3	2DS02_Curr	COD Currency	3		AN	1	F		ISO 4217 (eg. EUR)
4	2DS02_Amount	COD Amount	10		AN	1	F		Separator "." (point)
5	2DS02_CollectType	Collection Type	1		N	1	F		(0 = cash / 1 = crossed cheque/ 2 = Credit card)
		Format Envelope trailer	1		A	1			$^{R}_{S}$

3.3.5.7.4. *Custom format block*

Field No	Name	Description	Max Size	Subfield Max Size	Type	Priority	Field / Sub Field	Mgt Rul	Additional Description
1	2DS03_Header	Format Envelope Header	2		N	1	F		"07"
2	2DS03_FormatID	BLOCL-ID = "S03X"	4		AN	1	F	16 17	"X" (AN) is the version of the record from "0".
3	2DS03_CompInformation	Customs, Complete Information Flag	1		N	1			indicates whether more characters/data are needed than in 2D-Code printable (0=complete; 1=incomplete)
4	2DS03_RecVAT	Receivers VAT/PID/EORI No	20		AN	X	F	12	Necessary information for clearance at destination.
5	2DS03_SendVAT	Senders VAT / EORI number	20		AN	X	F	13	Senders VAT / EORI number (Economic Operator Registration & Identification)
6	2DS03_ComBillRecName	Commercial invoice receiver name	35		AN	X	F	14	Set in 20% of the case
7	2DS03_ComBillRecPropNum	Commercial invoice Property Number / House No	8		AN	X	F	14 6	Set in 20% of the case
8	2DS03_ComBillRecStreet	Commercial invoice receiver street	35		AN	X	F	14	Set in 20% of the case
9	2DS03_ComBillRecCity	Commercial invoice receiver city	25		AN	X	F	14	Set in 20% of the case
10	2DS03_ComBillRecCountryCode	Commercial invoice receiver country code	3		N	X	F	14 15	ISO 3166 (eg. 826 for GB), set in 20% of the case
11	2DS03_ComBillRecZipCode	Commercial invoice receiver postal code	9		AN	X	F	14	Set in 20% of the case
12	2DS03_ComBillRecContact	Commercial invoice receiver contact	35		AN	10	F	14	
13	2DS03_ComBillRecPhone	commercial invoice receiver phone number	25		AN	10	F	14	The only acceptable alphabetic character Is "+" before country code. Country code can be preceded of "00" or "+"
14	2DS03_TotalValue	Total Value	12		N	1	F		declared customs amount in total (complete shipment if MPS). Separator "." (point).
15	2DS03_Currency	Currency	3		A	1	F		ISO 4217 (e.g. EUR)
16	2DS03_Incoterm	Incoterm	3		A	1	F		DDU, DAP…. List incoterm 2010.
17	2DS03_DestCountryReg	Destination country registration information	15		AN	2	F		Registration number or FDA

Field No	Name	Description	Max Size	Subfield Max Size	Type	Priority	Field / Sub Field	Mgt Rul	Additional Description
18	2DS03_ArticleNumber	Number of article	3		N	1	F		Real number of article (could be more than five). If more than five, additional keyboard entry Is necessary
		Article description-1	45	66		2		2 18 30	
19	2DS03_Art1_Desc	- *Description*		35	AN		S		(medium value 17)
20	2DS03_Art1_Qty	- *Quantity*		2	N		S		
21	2DS03_Art1_Weigth	- *Weight*		4	N		S		Net weight in Decagram without separator
22	2DS03_Art1_Value	- *Value*		12	N		S		value of the invoice position. The currency Is the same currency than the total value. Separator "." (point).
23	2DS03_Atr1_ComCode	- *Commodity Code (NDP, HTC, HS)*		10	AN		S		Minimum of 6 digit. customs tariff number
24	2DS03_Art1_OriginCountry	- *Origin country code*		3	N		F		ISO-3166 3N (eg. 826 for GB), Manufactory country code.
25		Article description-2	45			2	F	2 18 30	
31		Article description-3	45			2	F	2 18 30	
37		Article description-4	45			2	F	2 18 30	
43		Article description-5	45			2	F	2 18 30	
		Format Envelope trailer	1		A	1			$^{R}{}_{S}$

3.3.5.7.5. *Return format block*

Field No	Name	Description	Max Size	Subfield Max Size	Type	Priority	Field / Sub Field	Mgt Rul	Additional Description
1	2DS04_Header	Format Envelope Header	2			1	F		"07"
2		BLOCL-ID = "S04X"	6				F		X (AN) is the version of the record from "0".
		Receiver Information	150	343		1		2 30	Note: The receiver Postal Code/Country is contained within the primary bar code
3	2DS04_RecComment	- *Receiver comment*		70	AN	5	S		Additional text to support delivery process. Any comment to help the delivery of the parcel. (Recipient comment)
4	2DS04_RecCompName2	- *Receiver Company Name2*		35	AN	10	S		Could be the company department (Communication department, finance...)
5	2DS04_RecContact	- *Receiver contact name*		35	AN	X	S	4	(receiver/consignee)
6	2DS04_RecContactPho1	- *Receiver Contact Phone Number1*		25	AN	X	S	25	The only acceptable alphabetic character Is "+" before country code. Country code can be preceded of "00" or "+"
7	2DS04_RecContactPho2	- *Receiver Contact Phone Number2*		25	AN	10	S		The only acceptable alphabetic character Is "+" before country code. Country code can be preceded of "00" or "+"
8	2DS04_RecNotifMob	- *Receiver Notification Mobile Phone number*		25	AN	X	S	5	The only acceptable alphabetic character Is "+" before country code. Country code can be preceded of "00" or "+"
9	2DS04_RecNotifEMail	- *Receiver Notification email*		50	AN	X	S	5	
10	2DS04_RecPropNum	- Receiver Property Number / House No		8	AN	2	S	6	
11	2DS04_RecAdd2	- *Receiver Address Line 2*		35	AN	2	S		
12	2DS04_RecAdd3	- *Receiver Address Line 3*		35	AN	2	F		
13	2DS04_NotifType	Notification type	1		A	X	F	26	Notification could be driven by this field or by the presence of the field notification phone/email (see rule number 5). "B" : Email+SMS, "E" : Email, "S" : SMS, "C" : Call, Empty : No notification.
14	2DS04_SOCode	Service code / Sorting Code	3		AN	1	F	32	DPDgroup Service Code / DPDgroup Sorting Code
15	2DS04_ASCode	Additional service code	6		AN	1	F	33	Three characters to start then up to 6 characters (RFU)
		Format Envelope trailer	1		A	1			$^{R}{}_{S}$

3.3.5.7.6. *Digital Label*

Field No	Name	Description	Max Size	Subfield Max Size	Type	Priority	Field / Sub Field	Mgt Rul	Additional Description
1	2DS05_Header	Format Envelope Header	2			1	F		"07"
2		BLOCL-ID = "S05X"	6				F		X (AN) is the version of the record from "0".
3	2DS05_Paid	Is the parcel allready paid?	1		A	1	F	34	"Y" for allready paid, "N" for not paid, "B" for to be bill. In this case, the filed "Customer Account Number" of the "ISO block" has to be set.
4	2DS05_Size	Parcel size	2		AN	C	F	35	Size of the parcel. Use to calculate the price of the transportation. To be check by the PUDO shop or other accredited people when the customer does the handover of the parcel. Could be: "XL","L","M","S","XS"
		Format Envelope trailer	1		A	1			$^{R}{}_{S}$

3.3.5.7.7. *Fresh/Food*

Field No	Name	Description	Max Size	Subfield Max Size	Type	Priority	Field / Sub Field	Mgt Rul	Additional Description
1	2DS06_Header	Format Envelope Header	2			1	F		"07"
2		BLOCL-ID = "S06X"	6				F		X (AN) is the version of the record from "0".
3	2DS06_GoodsType	Type of goods	3		N	1	F	36	Type of goods tramsported: 1 : Food 2 : Medics
4	2DS06_FeaturesCode	Feature of transportation	3		N	1	F	36	Type of Fresh service to be provided: 1 : Fresh (Range of temperature from 0°c up to 8°c) 2 : Frozen (Range of temperature under -18°c)
5	2DS06_ExpirationDate	best-before date	8		N	1	F	38	Limit date of consumption (best-before date)
		Format Envelope trailer	1		A	1			$^{R}{}_{S}$

3.3.5.7.8. *Fixed time delivery*

Field No	Name	Description	Max Size	Subfield Max Size	Type	Priority	Field / Sub Field	Mgt Rul	Additional Description
1	2DS07_Header	Format Envelope Header	2			1	F		"07"
2		BLOCL-ID = "S07X"	6				F		X (AN) is the version of the record from "0".
3	2DS07_DeliveryDateFrom	Ordered delivery date from	8		N	1	F	37	Format DDMMAAAA Specify that delivery can't be done before this date Can be empty if there is only a DeliveryDateTo
4	2DS07_DeliveryDateTo	Ordered delivery date to	8		N	1	F	37	Format DDMMAAAA Specify that delivery can't be done after this date Can be empty if there is only a DeliveryDateFrom
5	2DS07_DeliveryTimeFrom	Ordered delivery time from	5		AN	1	F	38	Delivery to be done from this time Specify that delivery can't be done before this time Can be empty if there is only a DeliveryTimeTo
6	2DS07_DeliveryTimeTo	Ordered delivery time to	5		AN	1	F	38	Delivery to be done up to this time Specify that delivery can't be done after this time Can be empty if there is only a DeliveryTimeFrom
		Format Envelope trailer	1		A	1			

3.3.5.7.9. *GPS Localization*

Field No	Name	Description	Max Size	Subfield Max Size	Type	Priority	Field / Sub Field	Mgt Rul	Additional Description
1	2DS08_Header	Format Envelope Header	2			1	F		"07"
2		BLOCL-ID = "S08X"	6				F		X (AN) is the version of the record from "0".
3	2DS08_Latitude	Latitude of the consignee location	10		AN	1	F		±DD.DDDD
4	2DS08_Longitude	Longitude of the consignee location	10		AN	1	F		±DDD.DDDD
5	2DS08_Quality	GPS encoding quality	1		N	1	F		Values: - "0" : Center of the street - "1" : Exact number in the street - "2" : Interpolation of the number in the street - "3" : Nearest number in the street - "9" : Unknown precision
6	2DS08_DataOrigin	Origin of the GPS data	1		N	1	F		Values: - "G" : GPS data produce by a DPDgroup software (or controled by DPDgroup) - Empty or other case
		Format Envelope trailer	1		A	1			^{R}S

3.3.5.8. *Physical Properties*

The specified physical properties of the DPDgroup AZTEC “2D” symbol have been developed to ensure accurate readability by the handheld scan guns in use now and with other bar code scanning systems that may be introduced in the future.

The individual components of the specification were determined as follows:

3.3.5.8.1. *Module size or X-Dimension*

Determined by the scanner type, the specified module size was selected to ensure adequate scanner performance with raster laser type scanners.

The width of the minimum narrow element (bar or space) is a dimensional parameter, which can be used to calculate the symbol’s actual width. This X-Dimension has to be a multiple of the size of a dot of our thermal printer. Most of our thermal printer are at least of 200 dpi (dot per inch). To optimize reading, X-Dimension should be at least equal to size of free dot. The X-Dimension should be set to 0,38mm

3.3.5.8.2. *Error Correction*

The error correction level was chosen as a good compromise between data security and symbol size (and overhead). ISO 24778 sets this error correction to 23% for large AZTEC barcode. ISO 24778 specify that that error correction should not be higher.

3.3.5.8.3. *Bar Code Metrics*

The important physical properties evaluated on the DPDgroup AZTEC “2D” symbol are:

Module Size:

X-Dimension (Module size)

Minimum:	N/A
Nominal:	0.38mm
Maximum:	N/A

Error Correction: 23%

Overall Bar Code Size:

Nominal Width:	3,4cm

Quiet Zones:

None

Label Stock:

Use only white label stock.
(Near Infrared for direct thermal printing)

3.3.5.9. *Print Quality*

The print quality specifications were developed to ensure all scanners could read the bar codes.

Print quality express in this chapter are in accordance with:

- ISO/IEC 24778 : “Information technology — Automatic identification and data capture techniques — Aztec Code bar code symbology specification”
- ISO/IEC 15415 : “Information technology — Automatic identification and data capture techniques — Bar code symbol print quality test specification — Two-dimensional symbols”

The major print quality criteria are:

3.3.5.9.1. *Print Contrast Signal (PCS)*

The Print Contrast Signal values were established to ensure the bar codes would be readable at the conveyor speeds used on the DPDgroup Ground automated package sorters. The scanners cannot consistently read bar codes with low PCS values on the sorters, although they may read well with a hand scanner. High quality white label stock along with high quality print is required to achieve the specified PCS. (Thermal labels should be of a Near Infra-Red type to support scanners operating in the 630nm to 670nm range. Labels should also exhibit stable print characteristics through exposure to varying heat and temperature conditions.)

3.3.5.9.2. *ISO 24778 Grade*

This is a graded measure of conformance to ISO standard for AZTEC print quality.

3.3.5.9.3. *AIM Uniform Symbology Specification Grade*

This is a graded measure of conformance to AIM (Automatic Identification Manufactures Association) standard for bar code print quality.

3.3.5.9.4. *Minimum Unused Error Correction*

Given as a percentage, this is a measure of

Print Contrast Signal

Minimum Print Contrast:	90%
Desirable Contrast Range:	95 – 100%

Symbol Contrast (min.): 65%

ISO 24778 Grade: “A” grade

AIM Uniform Symbology Specification Grade:
“A” grade

The overall print quality must be ANSI grade “A”.

3.4. Geolabel Formats

3.4.1. Label home/business delivery

Label used for Home or business delivery.

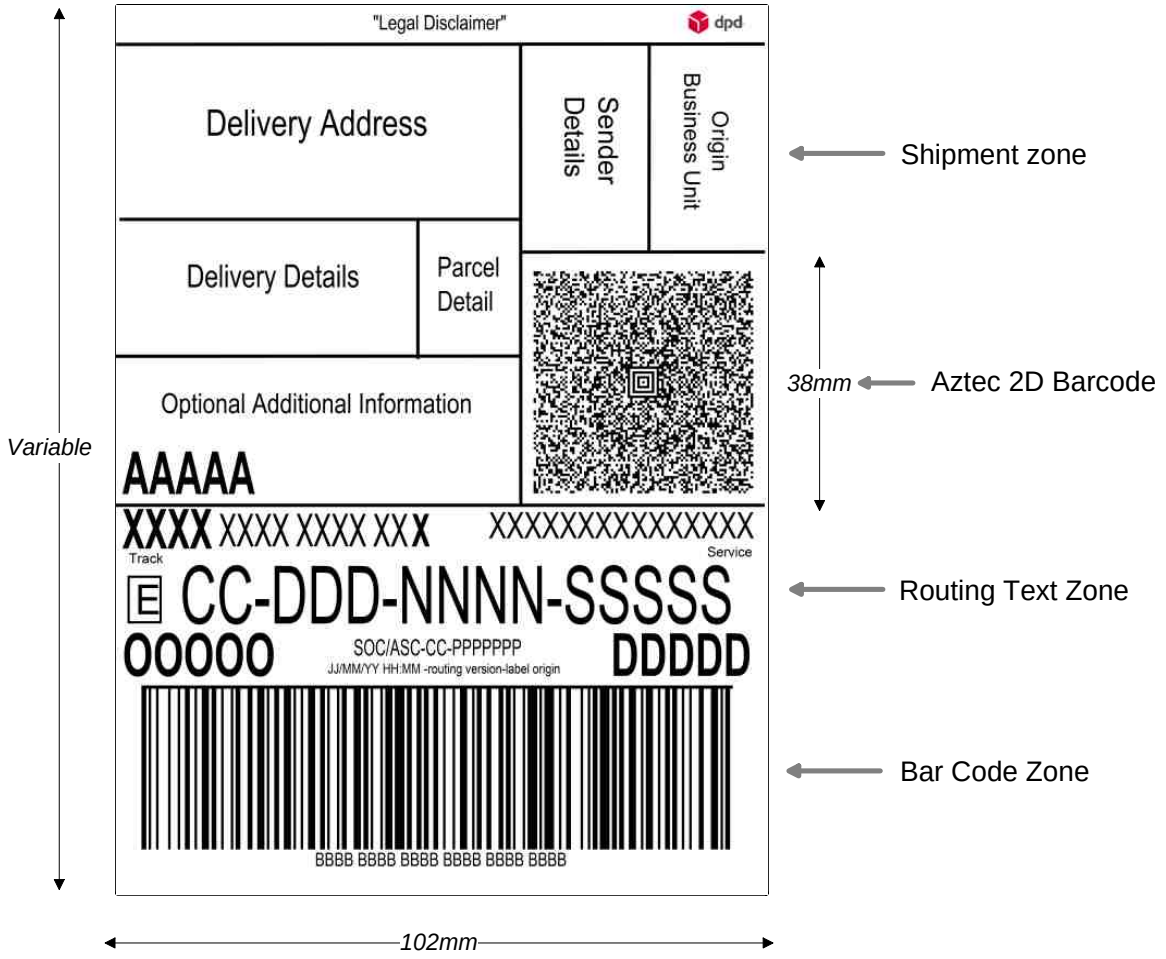
3.4.1.1. *Label Field Sizes*

Field Name	*Mandatory /Optional*	*Number of Characters*	*Text Height (20% Tolerance)*	*Comment*
ROUTING ZONE				
See Routing Label part definition				
HEADER ZONE				
Field Headings & Legal Text	M	Variable	1.5mm	
DELIVERY ADDRESSS ZONE				
Destination Company or name	M	30	2.5mm **bold**	
Destination address		30 (4 Lines)	2.5mm	
Destination country code	M	2	2.5mm **bold**	
Destination country	M	25	2.5mm **bold**	
DELIVERY DETAILS ZONE				
Destination Contact	O	20	2mm	
Destination Phone No	O	20	2mm	
Destination Info	O	20 (2 Lines)	2mm	
Consignment Number	O	16	2mm	
SENDER DETAILS ZONE				
Sender Account	M	10	1.5mm	
Sender Company or name	M	30	1.5mm **bold**	
Sender address		30 (4 Lines)	1.5mm	
Sender country code	M	2	1.5mm **bold**	
Sender country	M	25	1.5mm **bold**	
Sender Phone No	O	20	1.5mm	
Sender Reference	O	20	1.5mm	
ORIGIN BUSINESS UNIT ZONE				
Origin Business Unit	M	Variable	-	BU dependent
PARCEL DETAIL ZONE				
Packages	M	5	2.5mm	X of X
Weight	O	6	2.5mm	
OPTIONAL ADDITIONAL INFORMATION ZONE				
CSORT	O	6	2.5mm	BU dependent
Product detail	O	Variable	2.5mm	BU dependent

The text size is dependent on the font capabilities of the label printer so a 20% tolerance is allowed. The label text size proportions must be maintained to ensure visual compatibility. The specification label examples use “Arial Narrow” font.

To distinguish between the letter “O” and the number “0” , the number should be printed with a diagonal line, if possible – i.e. “Ø”

3.4.1.2. *Thermal AZTEC Label Format Home / Business delivery*



3.4.1.3. *Thermal Label AZTEC home delivery sample*

3.4.1.4. *Thermal Label AZTEC PUDO return delivery sample*

3.4.2. Label PUDO delivery

Label used for PUDO point delivery parcels.

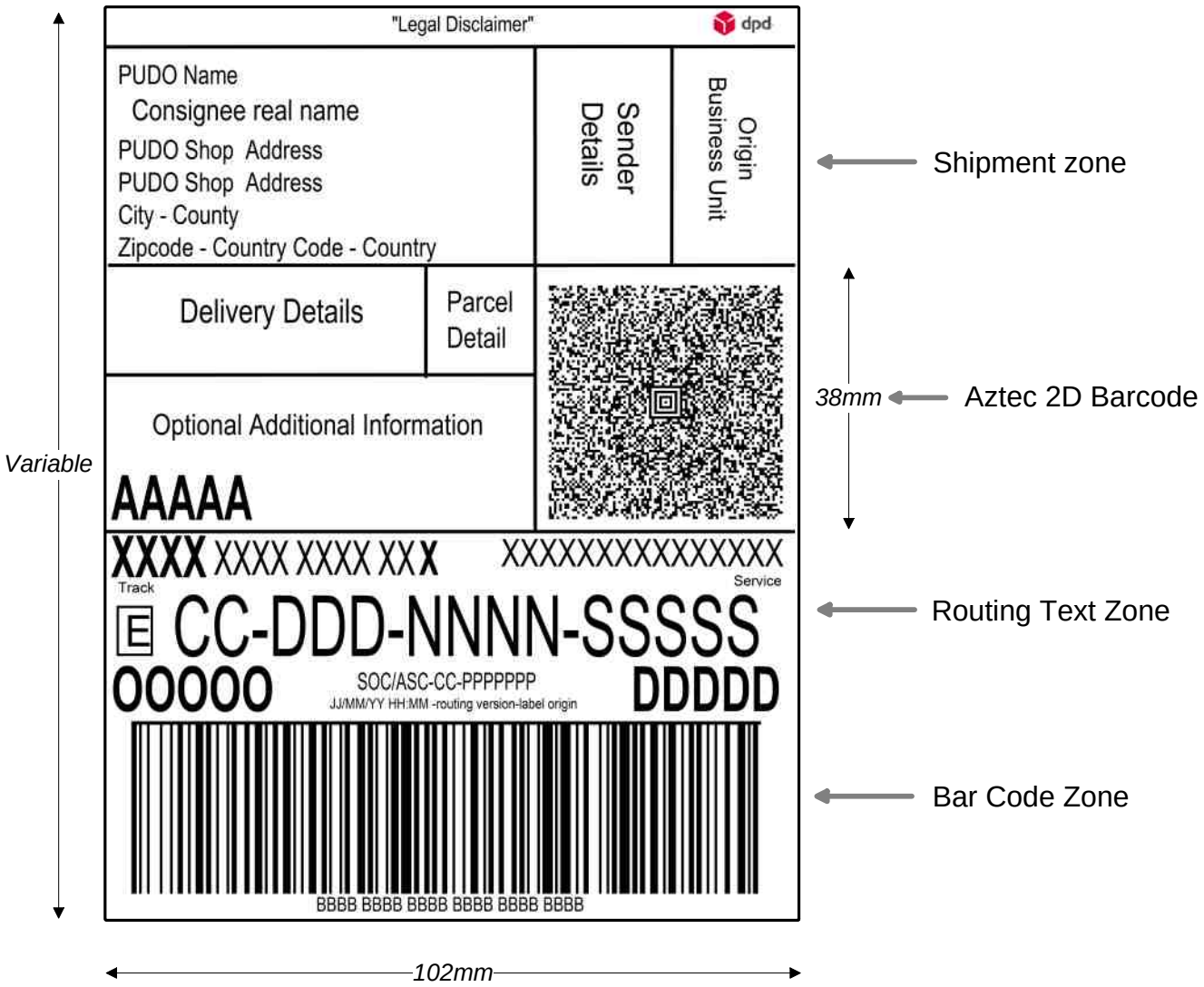
3.4.2.1. *Label Field Sizes*

Field Name	*Mandatory /Optional*	*Number of Characters*	*Text Height (20% Tolerance)*	*Comment*
ROUTING ZONE				
See Routing Label part definition				
HEADER ZONE				
Field Headings & Legal Text	M	Variable	1.5mm	
DESTINATION PUDO SHOP ADDRESSS ZONE				
PUDO shop name	M	30	2.5mm **bold**	
Real Consignee name	M	30	2.5mm **bold**	
Destination PUDO address	M	30 (2 Lines)	2.5mm	
Destination PUDO city+County	M	30	2.5mm **bold**	
Destination PUDO country code	M	2	2.5mm **bold**	
Destination PUDO country	M	25	2.5mm **bold**	
DELIVERY DETAILS ZONE				
Destination Contact	O	20	2mm	
Destination Phone No	O	20	2mm	
Consignment Number	O	16	2mm	
SENDER DETAILS ZONE				
Sender Account	M	10	1.5mm	
Sender Company or name	M	30	1.5mm **bold**	
Sender address		30 (4 Lines)	1.5mm	
Sender country code	M	2	1.5mm **bold**	
Sender country	M	25	1.5mm **bold**	
Sender Phone No	O	20	1.5mm	
Sender Reference	O	20	1.5mm	
ORIGIN BUSINESS UNIT ZONE				
Origin Business Unit	M	Variable	-	BU dependent
PARCEL DETAIL ZONE				
Packages	M	5	2.5mm	X of X
Weight	O	6	2.5mm	
OPTIONAL ADDITIONAL INFORMATION ZONE				
CSORT	O	6	2.5mm	BU dependent
Product detail	O	Variable	2.5mm	BU dependent

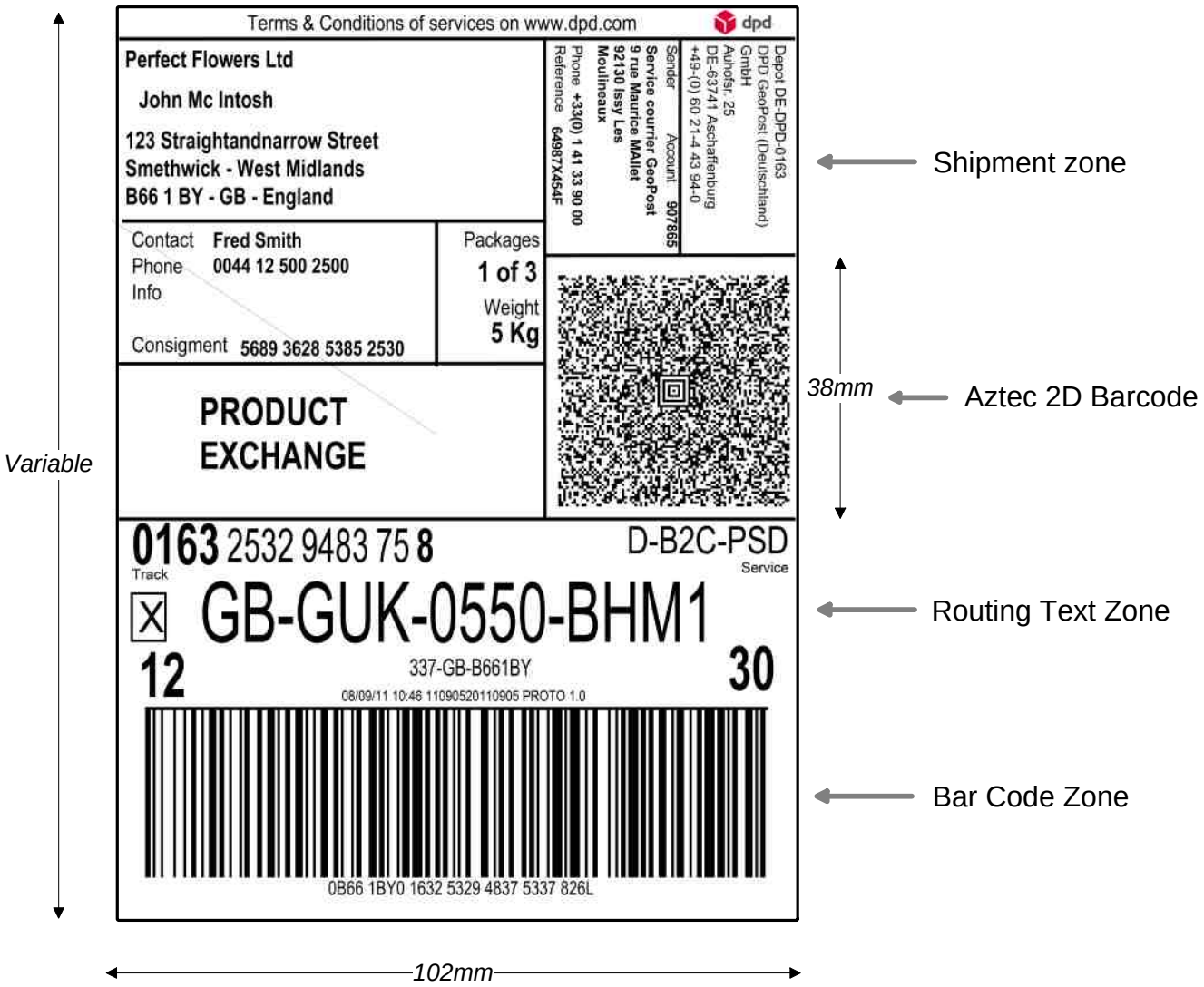
The text size is dependent on the font capabilities of the label printer so a 20% tolerance is allowed. The label text size proportions must be maintained to ensure visual compatibility. The specification label examples use “Arial Narrow” font.

To distinguish between the letter “O” and the number “0” , the number should be printed with a diagonal line, if possible – i.e. “Ø”

3.4.2.2. *Thermal AZTEC Label Format PUDO delivery*



3.4.2.3. *Thermal Label AZTEC PUDO delivery sample*



3.5. GeoLabel Validation Process

Each BU is responsible to check if the labels created by the shipper are in line with the Geolabel Specification.

The DPDgroup provides a Global Validation Center to validate GeoLabel V1.0 based Labels. The Global Validation Center will do a check based on the following topics:

- GeoLabel Format
- Mandatory Label Zones
- Routing Details Text& Sizes
- Code128 Spec &Print Quality
- Aztec 2D-Code Spec & Print Quality

It is mandatory to validate sample Labels generated by BU own shipping system. The Global Validation Center also accepts Labels generated by the shipper, but the BU is responsible for the communication with the shipper.
The full process of the validation Process is described in the “GeoLabel Validation Guideline”.

3.6. Impact of Geolabel Bar Code Quality on Sortation

3.6.1. Overview

DPDgroup relies on the latest technology in bar codes to sort and track packages, unlike any other small package transportation company. Packages are sorted at automated hubs at speeds in excess of 168 meters per minute. At these speeds, it is important that the bar code meet the DPDgroup specifications so the overhead scanners can read the bar code. If the bar code cannot be read, the package is routed to the relabel area. A duplicate of the original bar code is applied to the package to ensure the package will sort properly. The package is then re-circulated through the hub sortation system.

3.6.2. Benefits of Quality Bar Codes

- Ensures the best possible guaranteed service
- Provides real time information regarding the status of shipments
- Ensures data capture is accurate and complete
- Minimizes handling of packages which in turn reduces claims
- Improved shipment
- Ensures accurate billing

3.6.3. Why Relabels Occur

If a package is not labeled properly, it will potentially need to be relabeled. The major contributing factors that cause packages to be relabeled are as follows:

- bar code placement
- damaged bar code
- poor quality bar code

3.6.4. Bar code Placement

Bar codes should never be placed in the following places:

- Under straps
- Over an edge/corner/seam
- Placed on the small end of the package
- Under tape
- Under plastic

3.6.5. Damaged Bar Code

If a label is damaged because it is ripped, torn, wrinkled, wet or soiled, the package will have to be re-labeled.

3.6.6. Poor Quality Bar Code

The quality of a DPDgroup bar code plays a significant role. If the quality of the bar code is poor, it cannot be read by the sortation scanners and will need to be relabeled. The following are typical bar code quality problems:

- Print head problems or Split Bars
- Faded or Voids in the Print
- Incorrect data content
- Quiet zone too small
- Smudged bar code
- Toner fusion problem
- Wrinkle in the print
- Duplicate package numbers
- Print contrast too low (It must be 90% or higher)

3.6.7. Keys to Success

- Ensure that the front line parcel processing operator understands the importance of maintaining bar code quality
- Be proactive in solving bar code quality issues
- Work with DPDgroup for help in resolving of bar code quality issues
- Establish preventative maintenance programs on labeling system